Auditable Failure Memory for ROS 2 Navigation: Supplementary Materials & Verification Audit Details

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Document Overview

This document provides supplementary technical specifications, precondition integrity constraints, cryptographic audit procedures, source-code traceability mappings, and kinematic ledger decomposition tables accompanying the paper *Auditable Failure Memory for ROS 2 Navigation: An Exploratory Comparison with Spatial Caching*.

1 S1. Precondition Integrity & Doorway Clearance Details

The passage state evaluation in `src/doorway_evaluator.py` enforces strict sensor-TF alignment rules prior to emitting a clearance classification:

1.1 S1.1 Precondition Rules

1. **Finite Transforms**: TF translation vector $\mathbf{t} = [x, y, z]^T$ and yaw angle ψ must be finite real numbers ($-\infty < x, y, z, \psi < \infty$).
2. **Temporal Alignment**: Scan and TF timestamps must satisfy $|t_{\text{tf}} - t_{\text{scan}}| \leq 0.80\,\text{s}$.
3. **Data Freshness**: Current simulation clock t_{sim} must satisfy $t_{\text{sim}} - t_{\text{tf}} \leq 1.0\,\text{s}$ and $t_{\text{sim}} - t_{\text{scan}} \leq 1.5\,\text{s}$.
4. **Ray Projections**:
   - Number of finite laser scan ranges $N_{\text{valid}} > 0$.
   - Number of projected rays intersecting the doorway bounding volume $N_{\text{intersect}} > 0$.
5. **Costmap Availability**: When costmap subgrids are provided, total cells in the Region of Interest (ROI) $N_{\text{total_cells}} > 0$.

If any precondition fails, the evaluator emits State(B) = UNKNOWN and does not alter memory state.

1.2 S1.2 Bounding Volume & Threshold Parameters

- Doorway center: $x = 0.0\,\text{m}, y = 1.20\,\text{m}$.
- ROI bounding box: $x \in [-0.15, 0.15]\,\text{m}, y \in [0.95, 1.45]\,\text{m}$.
- Minimum pass-through rays for `FREE`: $N_{\text{pass_through}} \geq 8$.
- Maximum hits inside for `FREE`: $N_{\text{hits_inside}} = 0$.
- Hits threshold for `OCCUPIED`: $N_{\text{hits_inside}} \geq 5$ or $N_{\text{costmap_occupied_cells}} > 0$.

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2 S2. Cryptographic Replay Audit Engine

To avoid reliance on self-reported runtime metrics, the independent replay auditor (`scripts/replay_and_score_p2c.py`) re-scores every Gazebo simulation run offline:

2.1 S2.1 Replay Integrity Verification Steps

1. **Manifest Audit**: Verifies every serialized file against `checksums.sha256` to detect modifications relative to the recorded manifest (detecting file changes relative to the manifest, rather than asserting intrinsic authenticity of the original underlying sensor data).

2. **Trajectory Distance**: Computes traversed distance by summing Euclidean distances between consecutive odometry sample positions $\langle x_k, y_k \rangle_{k=1}^{K}$:

$$d = \sum_{k=2}^{K} \sqrt{(x_k - x_{k-1})^2 + (y_k - y_{k-1})^2} \tag{1}$$

3. **Halt Stability Verification**: Verifies that upon goal completion, the robot remains stationary over a minimum nominal window of 2.0 s:

$$\max_{t \in [t_{\text{goal}}, t_{\text{goal}} + \Delta t_{\text{stab}}]} |v_{\text{lin}}(t)| \leq 0.05\,\text{m/s}, \quad \max_{t \in [t_{\text{goal}}, t_{\text{goal}} + \Delta t_{\text{stab}}]} |v_{\text{ang}}(t)| \leq 0.08\,\text{rad/s} \tag{2}$$

4. **Chronological Causality**:

$$t_{\text{rec}} \geq \max(t_{\text{action_end}}, t_{\text{obs_eval}}) - 0.05\,\text{s} \tag{3}$$

$$t_{\text{inv}} \geq \max(t_{\text{failure}}, t_{\text{free_eval}}) - 0.05\,\text{s} \tag{4}$$

3 S3. Reproducible PDF Build Pipeline

The manuscript build pipeline (`paper/scripts/build_paper_pdf.py`):

1. **Dynamic Table Injection**: Substitutes table markers with freshly generated LaTeX and Markdown tables.

2. **MathJax Vector SVG Pre-Rendering**: Converts formulas into inline SVG vector paths via Node.js `mathjax-full@3.2.2`.

3. **Table Row Validation**: Verifies table row counts (Table 1: 10 data rows, Table 2: 10 data rows, Table 3: 4 data rows).

4. **Zero Raw TeX Leak Assertion**: Renders compiled PDF pages to PNG at 150 dpi and extracts text (`pdftotext`) to ensure zero unrendered TeX control sequences.

5. **Structured Build Record**: Emits `paper/build_report.json` documenting tool versions, git status, and cryptographic digests.

4 S4. Detailed Parameter & Source Code Traceability

For exact code symbol and line-number mappings for all mathematical formulations and protocol parameters, refer to `paper/method-code-map.csv` (33 parameter mappings). For the complete literature primary verification ledger, refer to `paper/reference-verification.csv` (15 primary-source verified references).

5 S5. D2 Trajectory Time Ledger: Observational Decomposition & Threshold Sensitivity

5.1 S5.1 Observational Trajectory Time Ledger & Algebraic Closure

To analyze timing differences between FailMem (F) and persistent suppression ($M1$) in Scenario D2 without relying on unverified stationary assumptions, we perform an observational trajectory decomposition across Full Episode, History Phase, and Decision Phase.

Notice on Algebraic Residuals: *The zero residual (residual $= 0.000\,s$) across all phases arises strictly by algebraic construction (absorbing unobserved active intervals and outside-action time as remainder components), and does not serve as independent validation of velocity classification accuracy or physical mechanism validity.*

Table 1: Decision Phase Ledger ($t \geq T_{\text{history}}$).

Condition	n	Distance (m)	Duration T_{dec} (s)	Moving T_{mov} (s)	Turning T_{turn} (s)	Stopped T_{stop} (s)	[Settling Window] (s)	Unobserved T_{unobs} (s)	Outside Actions (s)
D2_F	3	5.89 ± 0.02	51.10 ± 0.95	28.70 ± 1.18	4.20 ± 0.82	15.83 ± 2.42	[6.97]	1.33 ± 0.06	1.03 ± 0.15
D2_M1	3	7.89 ± 0.02	51.30 ± 2.00	34.30 ± 1.76	3.47 ± 0.49	12.13 ± 1.41	[4.60]	0.70 ± 0.00	0.70 ± 0.00
D2_O	3	5.94 ± 0.07	52.03 ± 1.75	29.33 ± 2.15	3.83 ± 0.49	16.47 ± 0.06	[6.93]	1.30 ± 0.10	1.10 ± 0.20
D2_R	3	5.92 ± 0.04	49.57 ± 1.59	28.27 ± 0.45	4.17 ± 0.84	14.70 ± 1.55	[5.37]	1.27 ± 0.06	1.17 ± 0.12

Table 2: History Phase Ledger ($t < T_{\text{history}}$).

Condition	n	Distance (m)	Duration T_{hist} (s)	Moving T_{mov} (s)	Turning T_{turn} (s)	Stopped T_{stop} (s)	[Settling Window] (s)	Unobserved T_{unobs} (s)	Outside Actions (s)
D2_F	3	10.36 ± 0.08	100.00 ± 7.72	52.37 ± 3.66	9.93 ± 0.75	30.13 ± 4.31	[11.00]	3.17 ± 0.15	4.40 ± 0.30
D2_M1	3	10.35 ± 0.11	97.53 ± 2.12	49.23 ± 1.58	10.10 ± 0.76	30.63 ± 0.91	[11.77]	3.20 ± 0.10	4.37 ± 0.31
D2_O	3	10.46 ± 0.22	97.73 ± 1.21	49.80 ± 1.14	9.83 ± 0.50	30.70 ± 0.78	[11.87]	3.23 ± 0.06	4.17 ± 0.42
D2_R	3	10.43 ± 0.05	98.60 ± 2.38	50.90 ± 1.32	9.40 ± 0.46	30.53 ± 1.80	[11.77]	3.07 ± 0.06	4.70 ± 0.27

Table 3: Full Episode Ledger.

Condition	n	Distance (m)	Duration T_{total} (s)	Moving T_{mov} (s)	Turning T_{turn} (s)	Stopped T_{stop} (s)	[Settling Window] (s)	Unobserved T_{unobs} (s)	Outside Actions (s)
D2_F	3	16.25 ± 0.10	151.10 ± 7.76	81.07 ± 4.48	14.13 ± 1.29	45.97 ± 2.57	[17.97]	4.50 ± 0.17	5.43 ± 0.45
D2_M1	3	18.25 ± 0.09	148.83 ± 2.40	83.53 ± 3.01	13.57 ± 0.51	42.77 ± 1.61	[16.37]	3.90 ± 0.10	5.07 ± 0.31
D2_O	3	16.40 ± 0.24	149.77 ± 0.55	79.13 ± 1.03	13.67 ± 0.15	47.17 ± 0.76	[18.80]	4.53 ± 0.12	5.27 ± 0.50
D2_R	3	16.34 ± 0.03	148.17 ± 3.16	79.17 ± 0.90	13.57 ± 1.21	45.23 ± 1.16	[17.13]	4.33 ± 0.06	5.87 ± 0.38

Note: Settling window duration (`sim_duration_covered`) is reported strictly as an independent event-level observation metric and is not algebraically subtracted from stopped time.

5.2 S5.2 Discretization Threshold Sensitivity Analysis

To evaluate sensitivity to kinematic classification parameters, Decision Phase moving and stopped durations were evaluated across linear velocity thresholds $v_{\text{lin}} \in \{0.03, 0.05, 0.08\}\,\text{m/s}$, angular velocity thresholds $\omega_{\text{ang}} \in \{0.05, 0.08, 0.12\}\,\text{rad/s}$, and gap thresholds $\tau_{\text{gap}} \in \{0.25, 0.50, 1.00\}\,\text{s}$.

Sensitivity Summary: Across thresholds, sample mean moving differences range from $-5.10\,\text{s}$ to $-6.80\,\text{s}$, and stopped differences range from $+3.10\,\text{s}$ to $+5.23\,\text{s}$. Because individual components depend on discretization thresholds, we report them as observational latencies without asserting exact physical compensation.

Table 4: Sensitivity of Moving and Stopped Durations to Discretization Thresholds.

v_{lin} (m/s)	ω_{ang} (rad/s)	τ_{gap} (s)	F Moving (s)	$M1$ Moving (s)	Diff (s)	F Stopped (s)	$M1$ Stopped (s)	Diff (s)
0.03	0.05	0.25	31.30	36.40	-5.10	13.80	10.70	+3.10
0.03	0.08	0.50	31.30	36.40	-5.10	13.80	10.70	+3.10
0.03	0.12	1.00	31.30	36.40	-5.10	13.80	10.70	+3.10
0.05	0.05	0.25	28.70	34.30	-5.60	15.77	12.07	+3.70
0.05	0.08	0.50	28.70	34.30	-5.60	15.83	12.13	+3.70
0.05	0.12	1.00	28.70	34.30	-5.60	15.90	12.13	+3.77
0.08	0.05	0.25	25.80	32.60	-6.80	18.33	13.13	+5.20
0.08	0.08	0.50	25.80	32.60	-6.80	18.47	13.27	+5.20
0.08	0.12	1.00	25.80	32.60	-6.80	18.53	13.30	+5.23